WASP-2b

R-0257

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_240820` · created 2026-07-17T07:53:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460542.8933 +/- 0.008 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.158 +/- 0.047
Transit depth (Rp/Rs)^2	2.51 +/- 1.48 [%]
Orbital Inclination (inc)	87.98 +/- 2.93
Airmass coefficient 1 (a1)	1.12 +/- 0.48
Airmass coefficient 2 (a2)	-0.21 +/- 0.16
Scatter in the residuals of the lightcurve fit is	38.5 %
Best Comparison Star	None
Optimal Aperture	4.88
Optimal Annulus	14.11
Transit Duration (day)	0.088 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.158 ± 0.047 vs 0.1326 (deviation 0.38σ)
Duration fit vs published	0.088 d vs 0.0737 d (deviation 0.56σ)
Mid-time O–C	20.5 min
Depth SNR	2.4
Residual scatter	38.5 %

Light curve

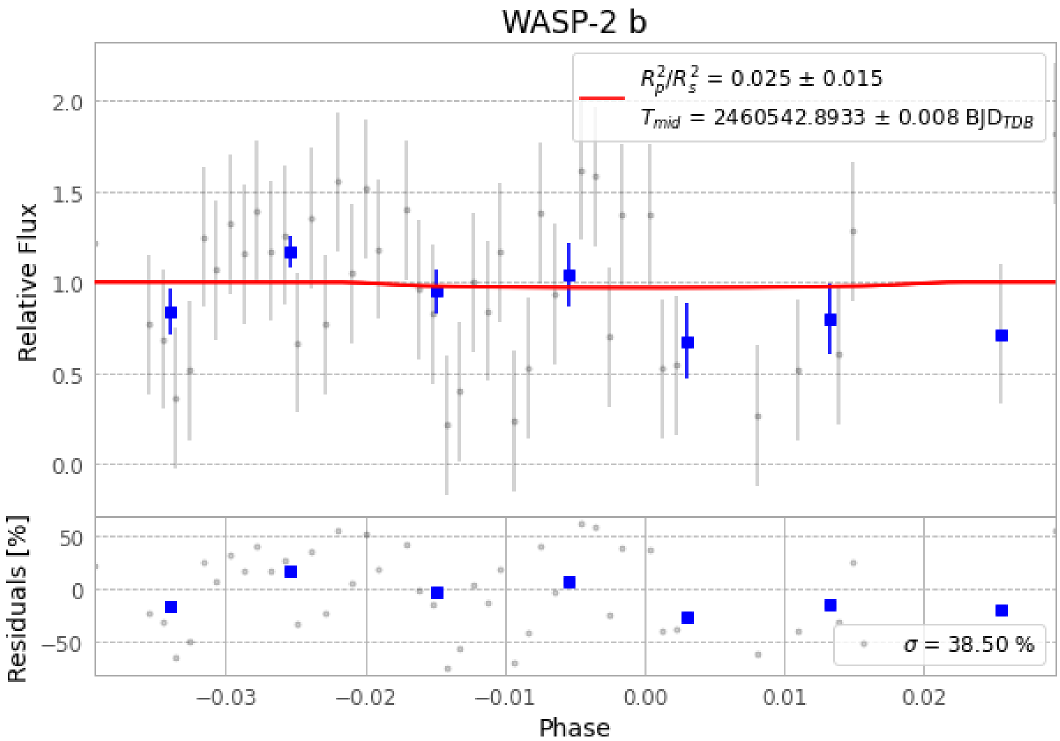

Figure 1 — FinalLightCurve_WASP-2 b_2024-08-20.png (SHA-256 d924177cfea0f9ba…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [370, 309], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 954ef4163f1328bb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 0eef8bc`

Data provenance

74 FITS frames (49 MB), `WASP-2240820071515.FITS` → `WASP-2240820105416.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-240820.log (dfa23941e70f…), FinalLightCurve_WASP-2 b_2024-08-20.png (d924177cfea0…), FinalLightCurve_WASP-2 b_2024-08-20.csv (d3b757dc8009…)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)